

human

What information do you have on wetlands in Exeter, plot them with clear markers

host_agent

Tool call: generate_metadata_for_all_artifacts - Arguments: {} - Response: {}

Tool call: send_message - Arguments: {'target': 'planning_agent', 'task_description': 'User wants to find and plot wetlands in Exeter with clear markers. What are the steps to solve this geospatial query?'}

planning_agent

The area specified is Exeter, so there will be 1 entry for Exeter. The entity to search for is wetlands, and the user wants to plot them with clear markers. There are no additional conditions specified, so I will focus on finding all wetlands in Exeter.

output steps: ["Find Exeter 1 area", "Search for wetlands in Exeter as many entries as possible", "Plot wetlands with clear markers"]

host_agent

Tool call: send_message - Arguments: {'target': 'named_area', 'task_description': 'Find the area polygon for Exeter.'}

named_area_agent

Tool call: call_os_ngd - Arguments: {'filters': None, 'bbox': None, 'polygon_or_point': True, 'street_address': None, 'filename': 'exeter_area_polygon'} - Response: Search results have been found. Artifact generated is : exeter_area_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter., count of records fetched : 98.

Tool call: send_message - Arguments: {'target': 'data_analysis_agent', 'task_description': "Filter the artifact 'exeter_area_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_area_polygon'."}

data_analysis_agent

Tool call: generate_metadata_for_artifacts - Arguments: {'artifact_names': ['exeter_area_polygon']}

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_area_polygon(data: list):
    gdf = data[0]
    # Filter using multiple columns to ensure we get the correct polygon for Exeter
    filtered = gdf[(gdf['name1_text'].str.lower() == 'exeter') |
                   (gdf['description'].str.contains('Exeter', case=False, na=False)) |
                   (gdf['theme'].str.contains('Exeter', case=False, na=False))]
    # Further filter to ensure high quality match
    filtered = filtered[filtered['description'].str.lower().str.contains('exeter', na=False) |
                       filtered['name1_text'].str.lower().str.contains('exeter', na=False)]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter."
    return [summary, 'filtered_exeter_area_polygon', 'GeoDataFrame containing only the polygon(s) for Exeter, filtered using name and description columns for h
```

Filtered to 1 polygon for Exeter. The results are saved as 'filtered_exeter_area_polygon', which contains only the polygon(s) for Exeter, filtered using both the name and description columns for high accuracy.

named_area_agent

The specific polygon for Exeter has been filtered and saved as 'filtered_exeter_area_polygon'. This GeoDataFrame contains only the polygon(s) for Exeter, ensuring high accuracy by filtering using both the name and description columns.

host_agent

Tool call: send_message - Arguments: {'target': 'land_features', 'task_description': "Search for wetlands within the 'filtered_exeter_area_polygon' artifact."}

land_features_agent

Tool call: call_os_ngd - Arguments: {'filters': ['Marsh', 'Marsh And Heath Or Rough Grassland', 'Marsh And Heath Or Rough Grassland And Scattered Coniferous Trees', 'Marsh And Heath Or Rough Grassland And Scattered Mixed Trees', 'Marsh And Heath Or Rough Grassland And Scattered Non-Coniferous Trees', 'Marsh And Heath Or Rough Grassland And Scrub', 'Marsh And Non-Coniferous Trees', 'Marsh And Non-Coniferous Trees And Scrub', 'Marsh And Scattered Boulders Or Scattered Rock', 'Marsh And Scrub', 'Saltmarsh'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'wetlands_filtered_exeter_area'} - Response: Multiple search results have been found in multiple datasets. Artifacts

generated are : ['land_wetlands_filtered_exeter_area'], Descriptions are : ["A geopandas dataframe containing land data with filters and bbox applied for the query Search for wetlands within the 'filtered_exeter_area_polygon' artifact. using filters ['Marsh', 'Marsh And Heath Or Rough Grassland', ...]. ", counts of records fetched are : [2].

The search for wetlands within the 'filtered_exeter_area_polygon' artifact has been completed. Here are the results:

- **Artifact Name:** land_wetlands_filtered_exeter_area
- **Description:** A geopandas dataframe containing land data with filters applied for wetlands, including marshes and saltmarshes.
- **Number of Records Fetched:** 2

If you need further analysis on this data, please let me know!

host_agent

Tool call: send_message - Arguments: {'target': 'land_use_features', 'task_description': "Search for wetlands within the 'filtered_exeter_area_polygon' artifact."}

land_use_features_agent

Tool call: call_os_ngd - Arguments: {'filters': ['Wetland'], 'bbox': 'filtered_exeter_area_polygon', 'polygon_or_point': True, 'street_address': False, 'filename': 'wetlands_in_exeter'} - Response: No results found for your search terms

No wetlands were found within the 'filtered_exeter_area_polygon' artifact.

host_agent

Tool call: send_message - Arguments: {'target': 'plotting_agent', 'task_description': "Plot the wetlands from the 'land_wetlands_filtered_exeter_area' artifact with clear markers, and overlay the 'filtered_exeter_area_polygon' to show the boundary of Exeter."}

plotting_agent

Tool call: generate_metadata_for_artifacts - Arguments: {'artifact_names': ['land_wetlands_filtered_exeter_area', 'filtered_exeter_area_polygon']}

Tool Code Executor

```

import matplotlib
matplotlib.use('Agg')
import folium
from shapely.geometry import mapping

def plot_wetlands_and_boundary(data: list):
    wetlands = data[0].copy()
    boundary = data[1].copy()

    # Ensure CRS is EPSG:4326 for folium
    if hasattr(wetlands, 'crs') and wetlands.crs is not None and wetlands.crs.to_epsg() != 4326:
        wetlands = wetlands.to_crs(epsg=4326)
    if hasattr(boundary, 'crs') and boundary.crs is not None and boundary.crs.to_epsg() != 4326:
        boundary = boundary.to_crs(epsg=4326)

    # Get centroid of boundary for map centering
    centroid = boundary.geometry.unary_union.centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13)

    # Plot wetlands polygons with clear color
    for _, row in wetlands.iterrows():
        geom = row['geometry']
        folium.GeoJson(
            mapping(geom),
            style_function=lambda x: {
                'fillColor': '#1f78b4',
                'color': '#08519c',
                'weight': 2,
                'fillOpacity': 0.5
            },
            tooltip=str(row['description']) if 'description' in wetlands.columns else 'Wetland'
        ).add_to(m)

    # Overlay Exeter boundary
    for _, row in boundary.iterrows():
        folium.GeoJson(
            mapping(row['geometry']),
            style_function=lambda x: {
                'fillColor': 'none',
                'color': 'red',
                'weight': 3,
                'fillOpacity': 0.1
            },
            tooltip='Exeter Boundary'
        ).add_to(m)

    # Save map
    map_filename = 'wetlands_exeter_map.html'

```

```
m.save(map_filename)

# Prepare summary
wetlands_summary = wetlands[['description', 'geometry_area_m2']].head(5)
summary = f"Plotted {len(wetlands)} wetlands in Exeter area. First 5 entries:\n" + str(wetlands_summary)
return [summary, 'wetlands_exeter_map', 'Folium map showing wetlands in Exeter area with city boundary overlay.', map_filename]
```

Plotted 2 wetlands in the Exeter area. First 5 entries: description geometry_area_m2 0 Marsh And Scrub 3495.752 1 Saltmarsh 5616.784

Artifact name: wetlands_exeter_map

Description: Folium map showing wetlands in Exeter area with city boundary overlay.

Map file: wetlands_exeter_map.html

host_agent

Here is the information found on wetlands in Exeter:

- Two wetlands were identified within the Exeter area: one classified as "Marsh And Scrub" and the other as "Saltmarsh."
- These wetlands have been plotted on a map with clear markers, and the boundary of Exeter is overlaid for context.

If you need the map file or further details about these wetlands, please let me know!